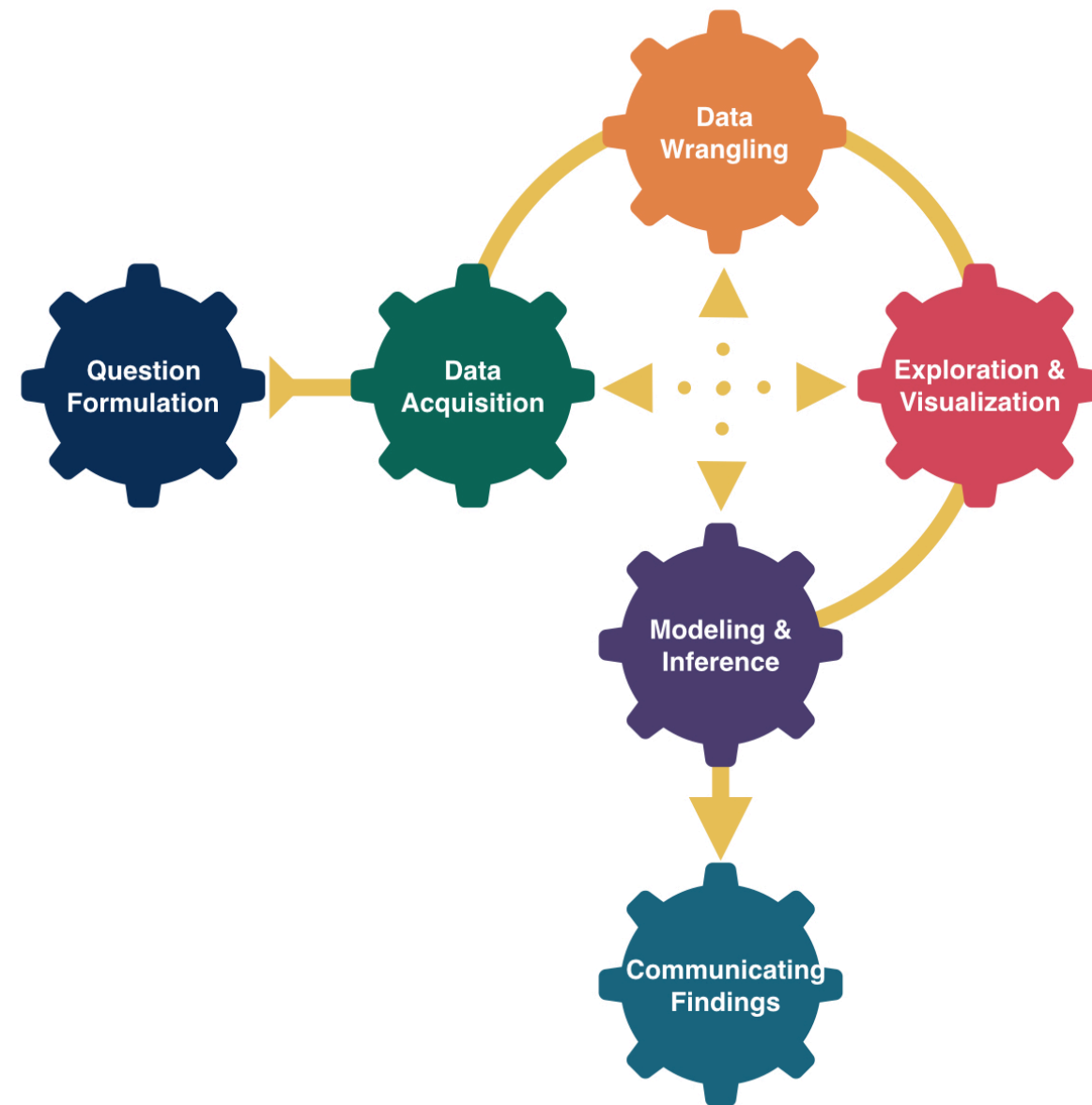


Modeling

Kelly McConville

DATA 100

Week 6 | Spring 2026



Announcements

- Exam grading in-progress
- Back to lab on Friday and P-Set 4.

Goals for Today

- Quick survey
- Modeling
- Measuring linear relationships

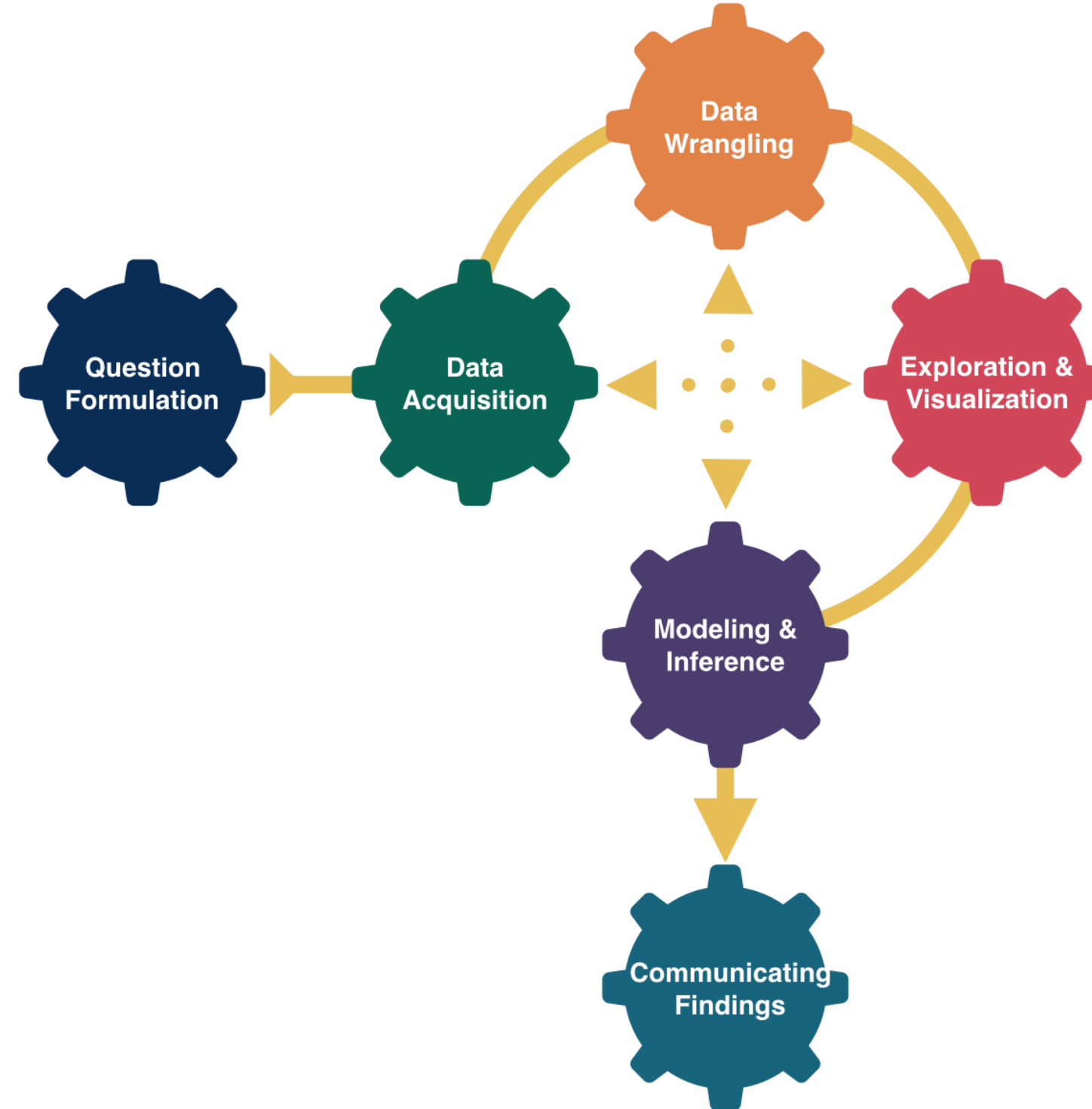
Reflecting on your DATA 100 Engagement

If you felt like the midterm didn't go as well as you would have hoped, I'd encourage you to reflect on how you are engaging with the DATA 100 support systems:

- Are you attending lecture?
- Are you attending lab and actively engaging with the material?
- Are you able to make progress on the p-sets by yourself before collaborating with others?
 - When collaborating, are you taking on an active or a passive role?
- Are you using office hours to talk through the material and your p-set?

Rebecca and I want to see you have a positive, productive learning experience in DATA 100!

Recap



Form of the Model

$$y = f(x) + \epsilon$$

Goal:

- Determine a reasonable form for $f()$. (Ex: Line, curve, ...)
- Estimate $f()$ with $\hat{f}()$ using the data.
- Generate predicted values: $\hat{y} = \hat{f}(x)$.

Simple Linear Regression Model

Consider this model when:

- Response variable (y): quantitative
- Explanatory variable (x): quantitative
 - Have only ONE explanatory variable.
- AND, $f()$ can be approximated by a line.

Example: The Ultimate Halloween Candy Power Ranking

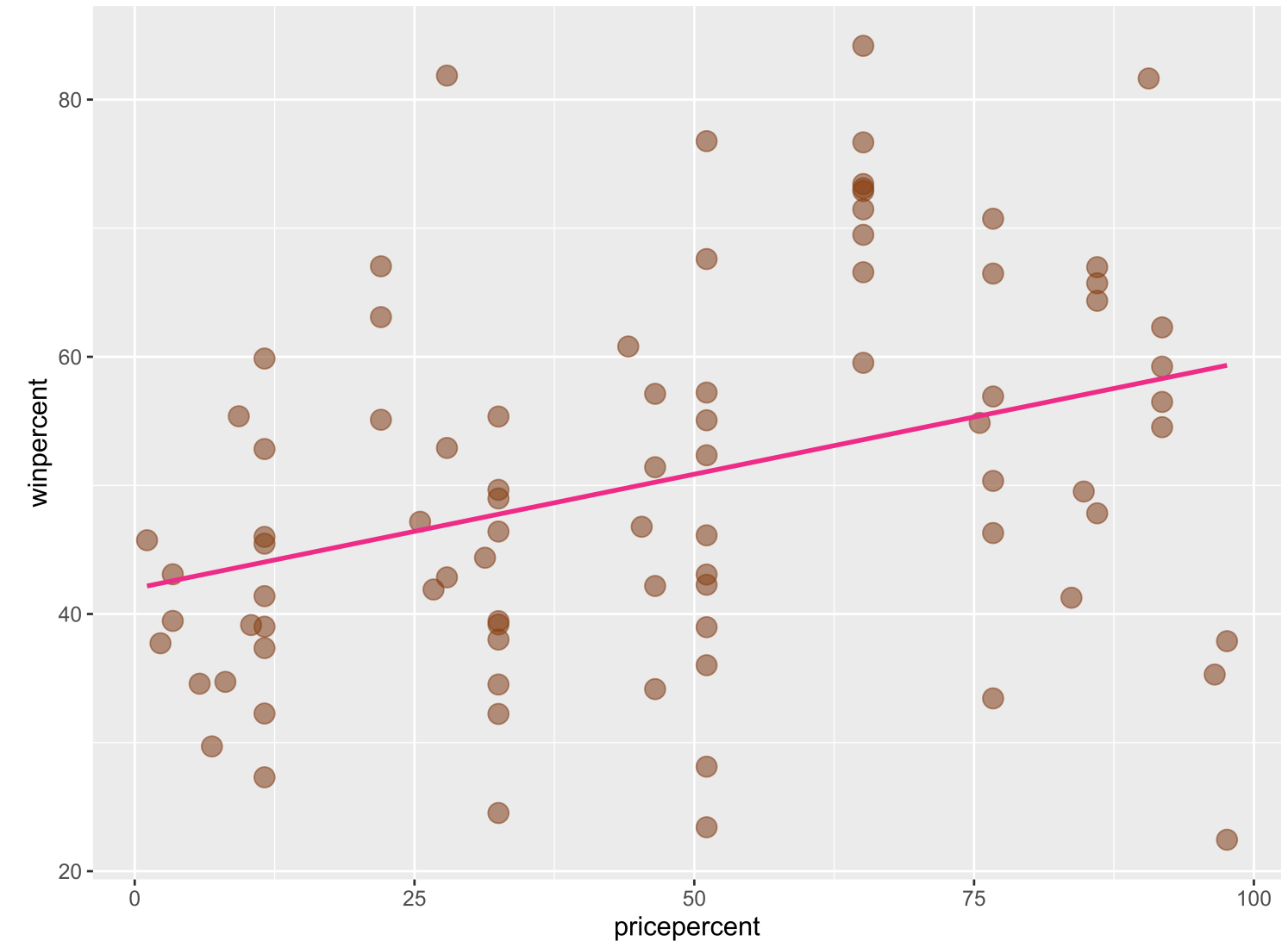
```
1 candy <- read_csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-ranking/candy-data.csv")
2   mutate(pricepercent = pricepercent*100)
3
4 glimpse(candy)
```

Rows: 85
Columns: 13

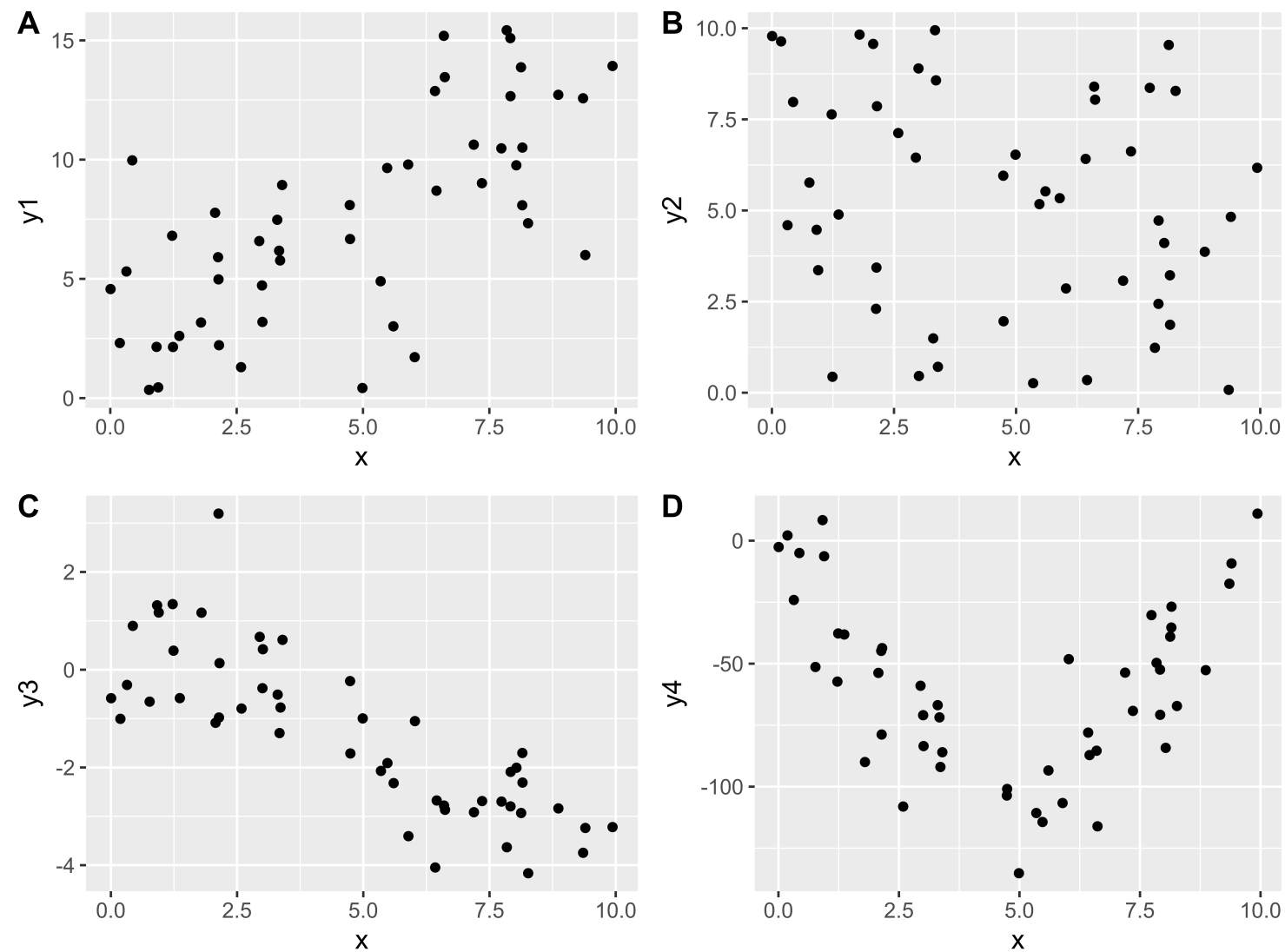
\$ competitorname	<chr>	"100 Grand", "3 Musketeers", "One dime", "One quarter...
\$ chocolate	<dbl>	1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,...
\$ fruity	<dbl>	0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1,...
\$ caramel	<dbl>	1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0,...
\$ peanutyalmondy	<dbl>	0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
\$ nougat	<dbl>	0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,...
\$ crispedricewafer	<dbl>	1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
\$ hard	<dbl>	0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1,...
\$ bar	<dbl>	1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,...
\$ pluribus	<dbl>	0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 0, 1,...
\$ sugarpercent	<dbl>	0.732, 0.604, 0.011, 0.011, 0.906, 0.465, 0.604, 0.31...
\$ pricepercent	<dbl>	86.0, 51.1, 11.6, 51.1, 51.1, 76.7, 76.7, 51.1, 32.5,...
\$ winpercent	<dbl>	66.97173, 67.60294, 32.26109, 46.11650, 52.34146, 50....

Example: The Ultimate Halloween Candy Power Ranking

- A simple linear regression model would be suitable for these data.
- But first, let's describe more plots!



Describing Relationships



- Need a summary statistics that quantifies the strength and relationship of the linear trend!

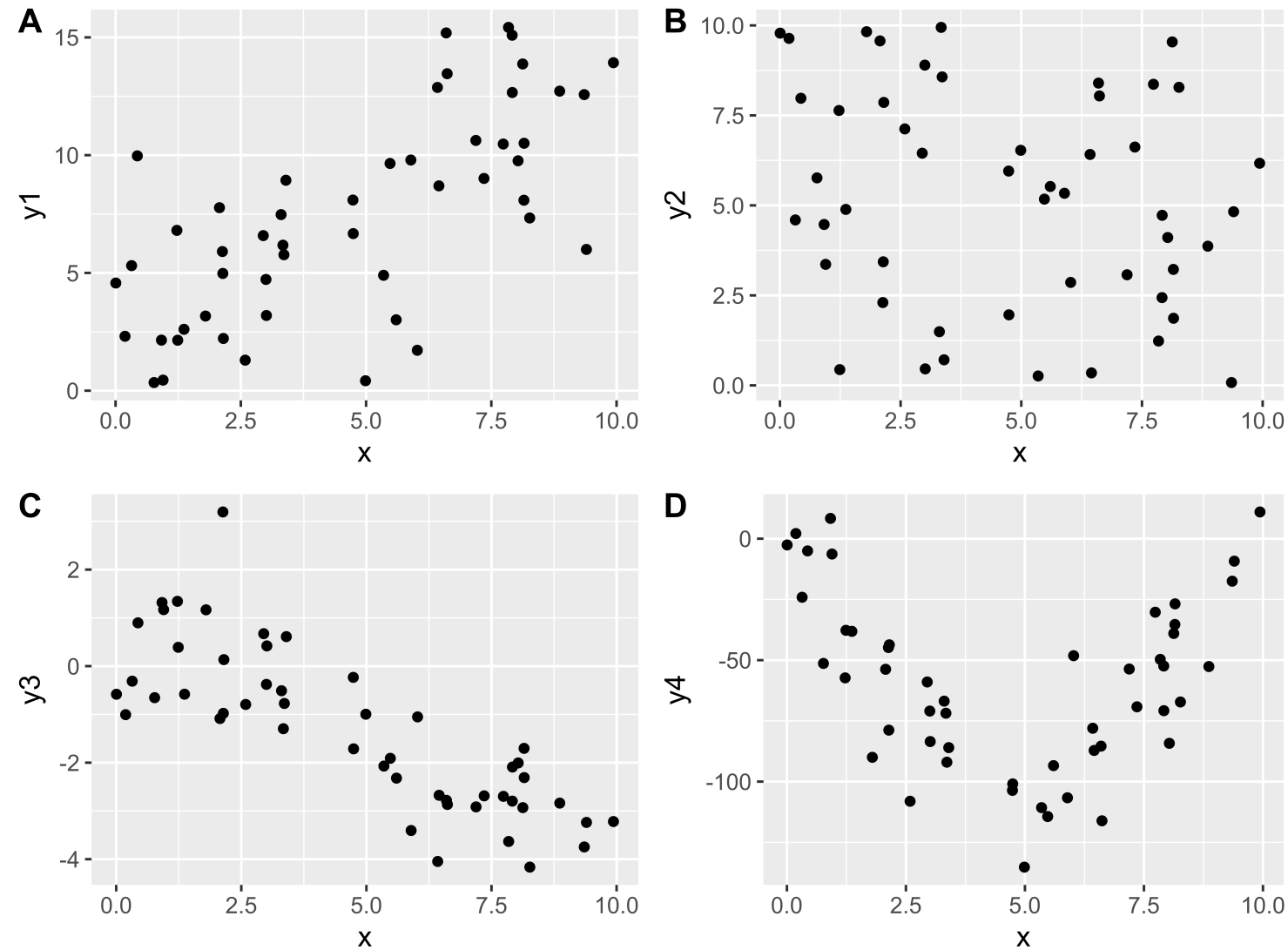
(Sample) Correlation Coefficient

- Measures the **strength** and **direction** of **linear** relationship between two quantitative variables
- Symbol: r
- Always between -1 and 1
- Sign indicates the direction of the relationship
- Magnitude indicates the strength of the linear relationship

```
1 candy %>%  
2   summarize(cor = cor(pricepercent, winpercent))  
  
# A tibble: 1 × 1  
  cor  
  <dbl>  
1 0.345
```


Any guesses on the correlations for A, B, C, or D?

```
1 dat %>%  
2   summarize(A = cor(x, y1), B = cor(x, y2),  
3             C = cor(x, y3), D = cor(x, y4))
```



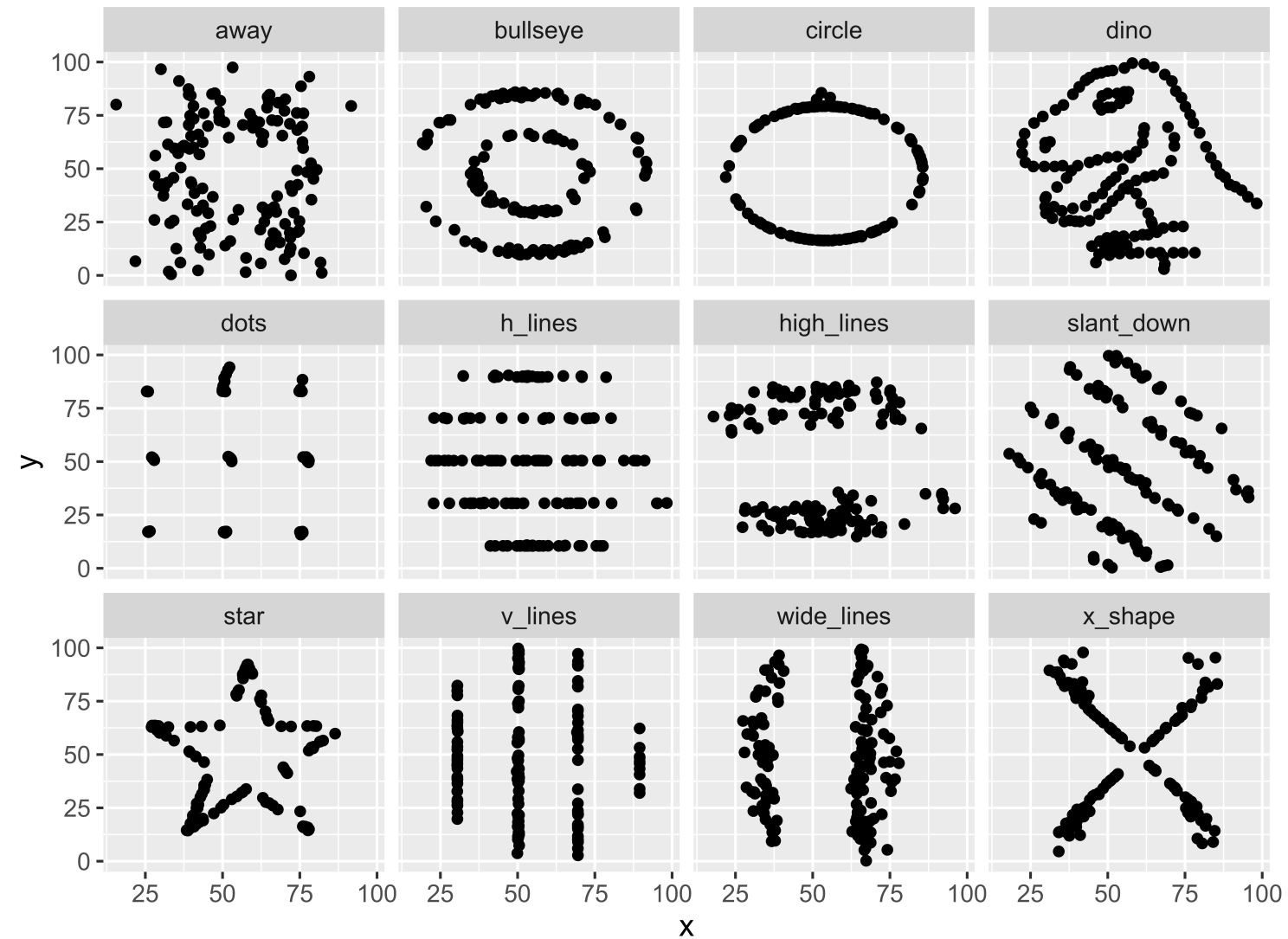
New Example

```
1 # Correlation coefficients
2 dat2 %>%
3   group_by(dataset) %>%
4   summarize(cor = cor(x, y))
```

```
# A tibble: 12 × 2
  dataset      cor
  <chr>      <dbl>
1 away      -0.0641
2 bullseye  -0.0686
3 circle    -0.0683
4 dino      -0.0645
5 dots      -0.0603
6 h_lines   -0.0617
7 high_lines -0.0685
8 slant_down -0.0690
9 star      -0.0630
10 v_lines  -0.0694
11 wide_lines -0.0666
12 x_shape  -0.0656
```

- Conclude that x and y have the same relationship across these different datasets because the correlation is the same?

Always graph the data when exploring relationships!



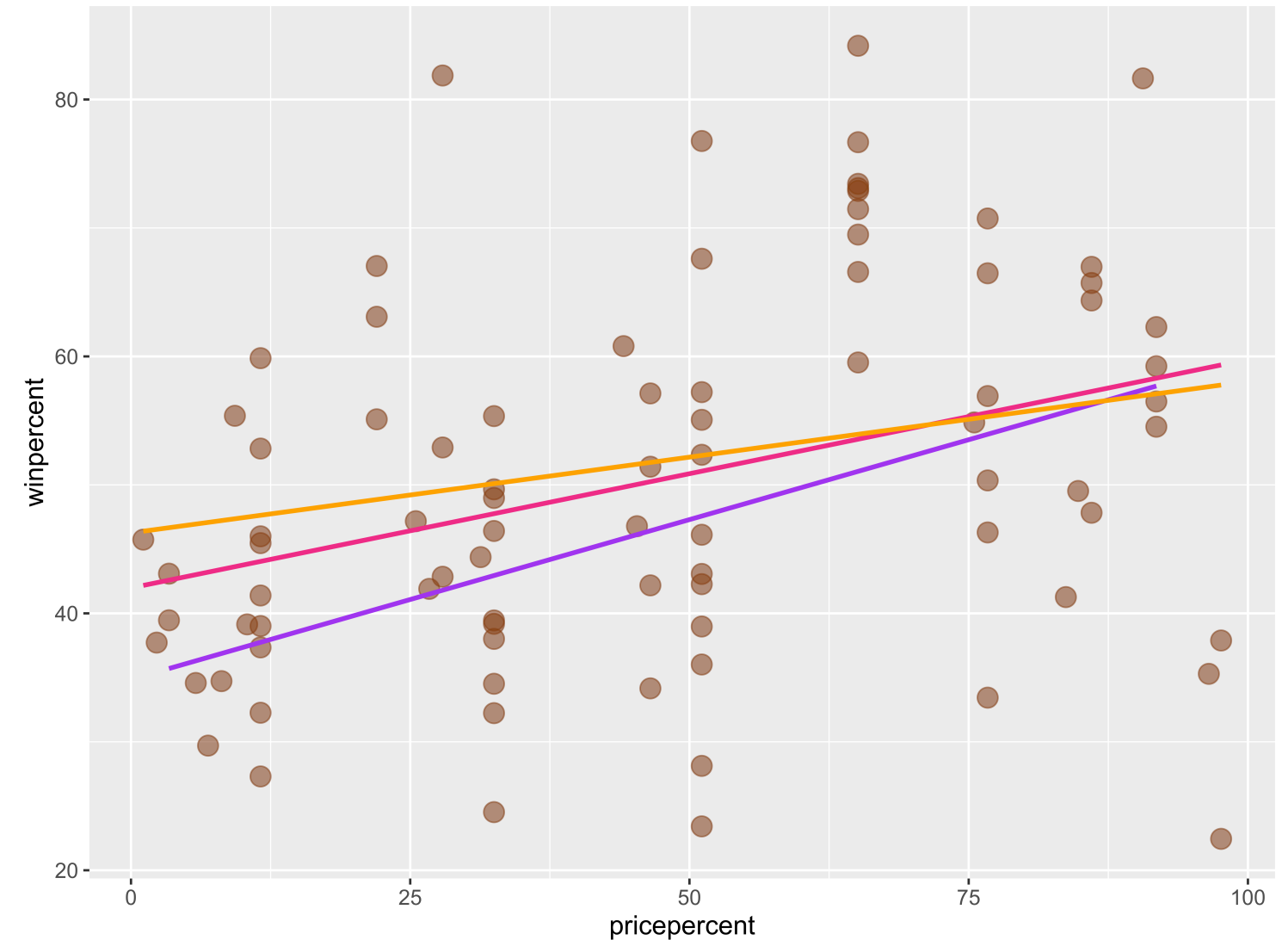
The datasets may have the same value for the correlation coefficient, but they don't exhibit the same relationships!

Returning to the Simple Linear Regression model...

Simple Linear Regression

Let's return to the Candy Example.

- A line is a reasonable model form.
- Where should the line be?
 - Slope? Intercept?



Form of the SLR Model

$$y = f(x) + \epsilon$$

$$y = \beta_0 + \beta_1 x + \epsilon$$

- Need to determine the best **estimates** of β_0 and β_1 .

Distinguishing between the **population** and the **sample**

$$y = \beta_0 + \beta_1 x + \epsilon$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- Parameters:
 - Based on the **population**
 - Unknown then if don't have data on the whole population
 - EX: β_0 and β_1
- Statistics:
 - Based on the **sample** data
 - Known
 - Usually estimate a population parameter
 - EX: $\hat{\beta}_0$ and $\hat{\beta}_1$

Method of Least Squares

Need two key definitions:

- **Fitted value:** The *estimated* value of the i -th case

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- **Residuals:** The *observed* error term for the i -th case

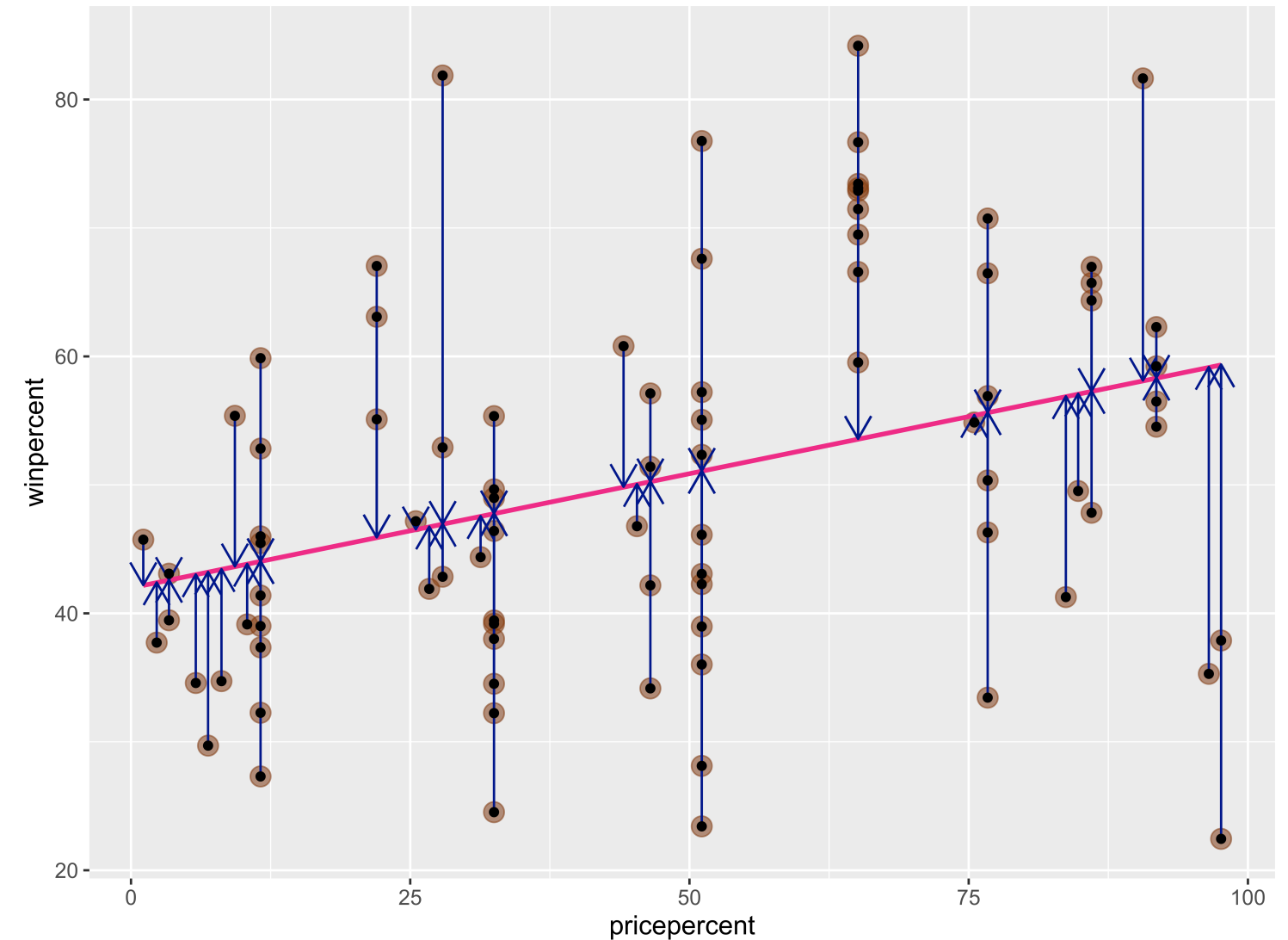
$$e_i = y_i - \hat{y}_i$$

Goal: Pick values for $\hat{\beta}_0$ and $\hat{\beta}_1$ so that the residuals are small!

Method of Least Squares

- Want residuals to be small.
- Minimize a function of the residuals.
- Minimize:

$$\sum_{i=1}^n e_i^2$$



Method of Least Squares

After minimizing the sum of squared residuals, you get the following equations:

Get the following equations:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad \text{and} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Method of Least Squares

Then we can estimate the whole function with:

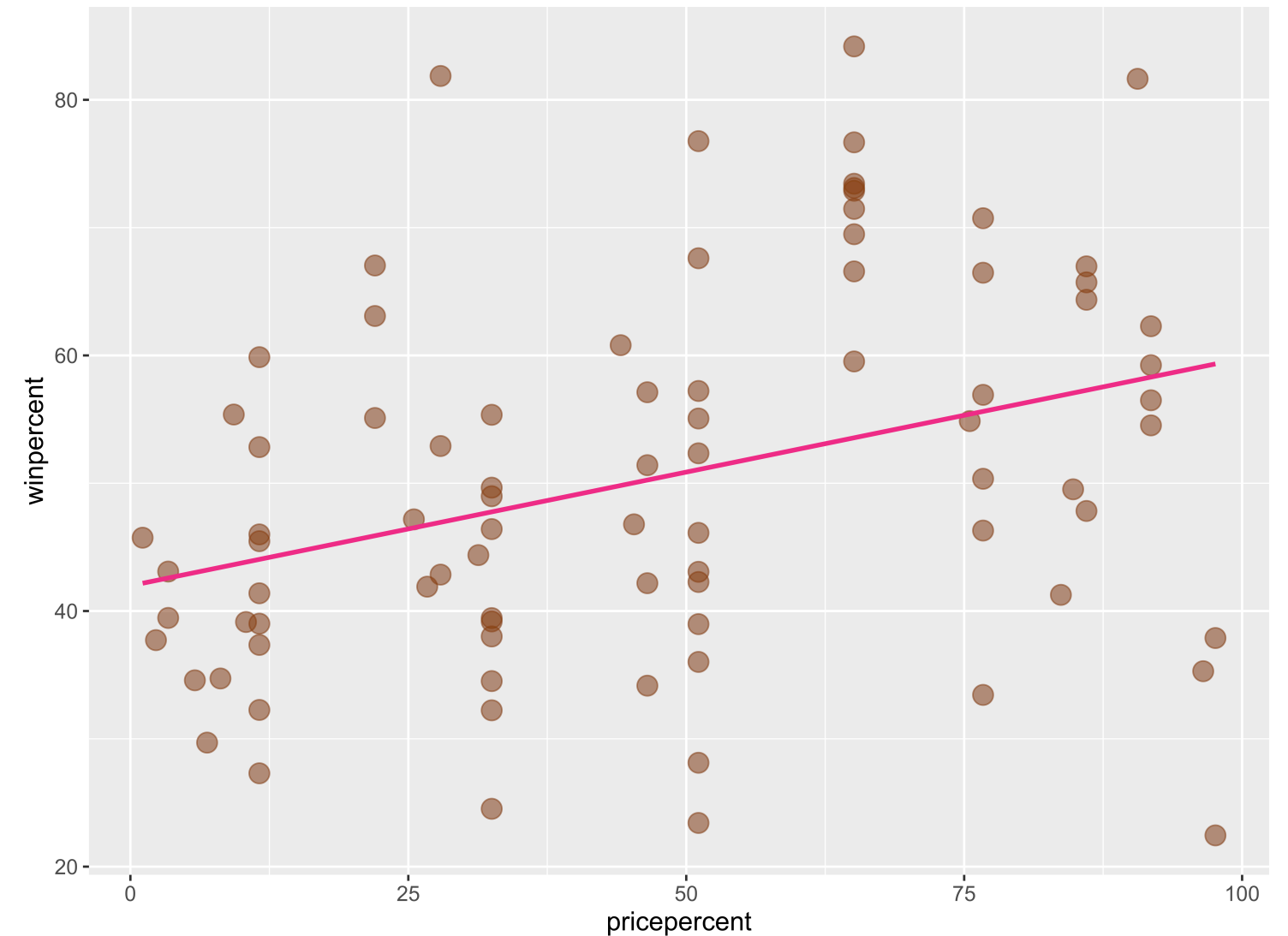
$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Called the **least squares line** or the **line of best fit**.

Method of Least Squares

`ggplot2` will compute the line and add it to your plot using `geom_smooth(method = "lm")`

```
1 ggplot(data = candy,  
2       mapping = aes(x = pricepercent,  
3                     y = winpercent)) +  
4   geom_point(alpha = 0.6, size = 4,  
5             color = "chocolate4") +  
6   geom_smooth(method = "lm", se = FALSE,  
7             color = "deeppink2")
```



But what are the **exact** values of $\hat{\beta}_0$ and $\hat{\beta}_1$?

Constructing the Simple Linear Regression Model in R

```
1 mod <- lm(winpercent ~ pricepercent, data = candy)
2
3 library(moderndive)
4 get_regression_table(mod)
```

A tibble: 2 × 7

	term	estimate	std_error	statistic	p_value	lower_ci	upper_ci
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	intercept	42.0	2.91	14.4	0	36.2	47.8
2	pricepercent	0.178	0.053	3.35	0.001	0.072	0.283

- Interpretation of the coefficients?

Prediction

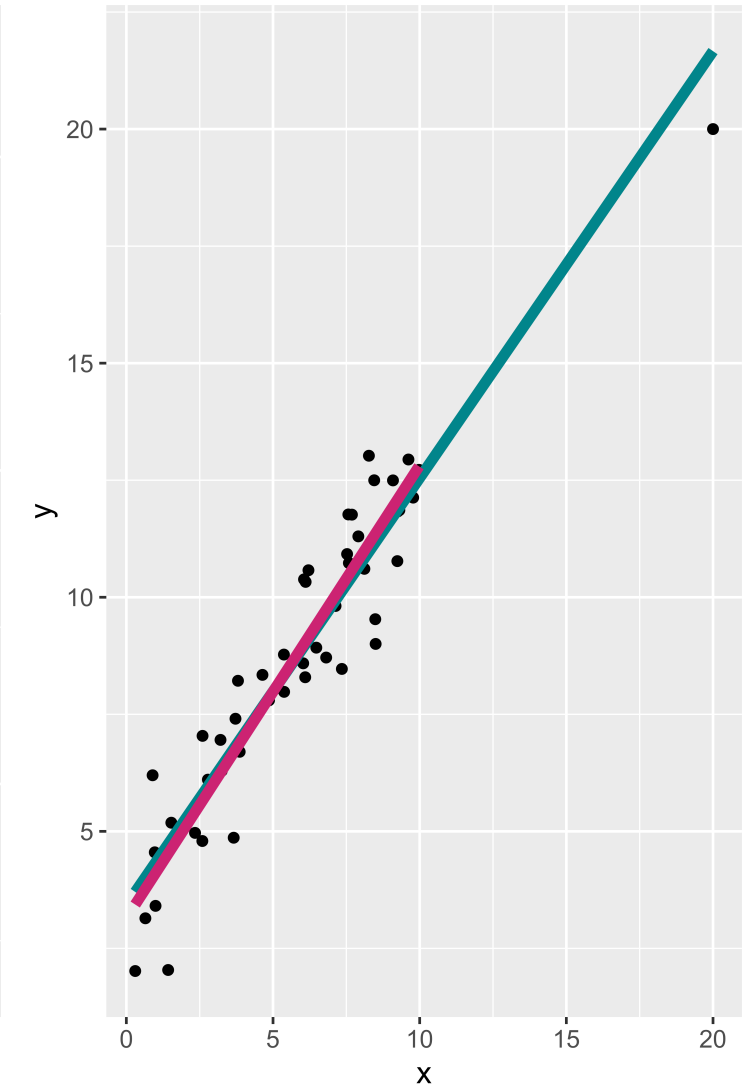
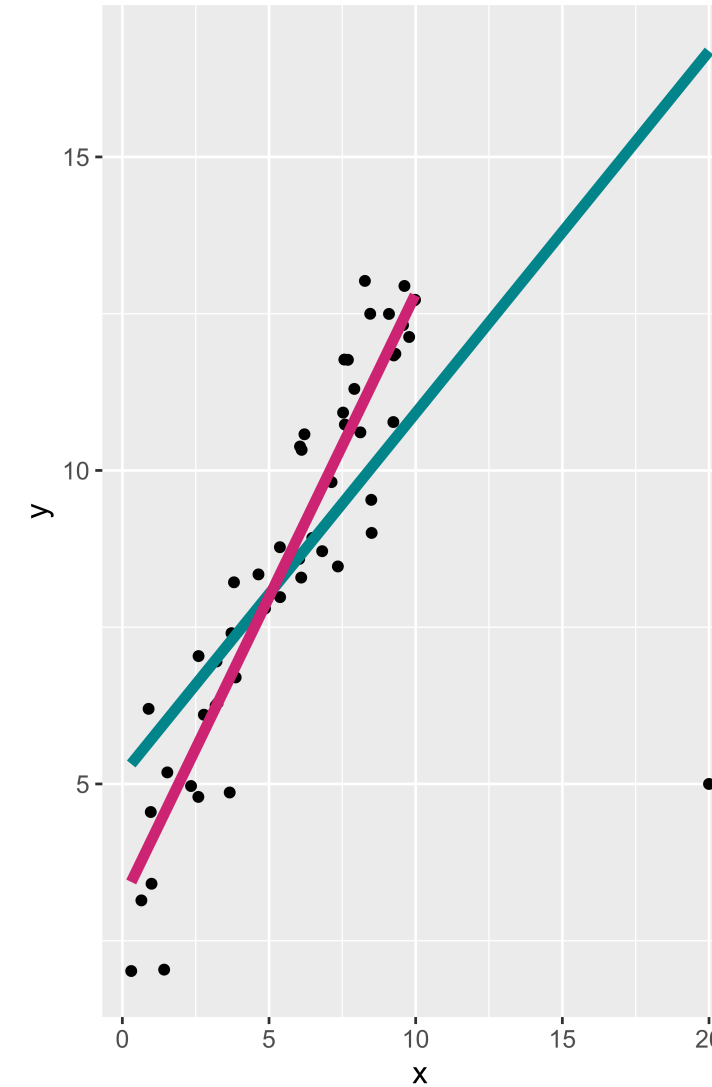
```
1 new_cases <- data.frame(pricepercent = c(25, 85, 150))
2 predict(mod, newdata = new_cases)
```

```
      1      2      3
46.42443 57.09409 68.65289
```

- We didn't have any treats in our sample with a price percentage of 85%. Can we still make this prediction?
 - Called **interpolation**
- We didn't have any treats in our sample with a price percentage of 150%. Can we still make this prediction?
 - Called **extrapolation**

Cautions

- Careful to only predict values within the range of x values in the sample.
- Make sure to investigate **outliers**: observations that fall far from the cloud of points.



Linear Regression

Linear regression is a flexible class of models that allow for:

- Both quantitative and categorical **explanatory** variables.
- **Multiple** explanatory variables.
- **Curved** relationships between the response variable and the explanatory variable.
- BUT the **response variable is quantitative**.

What About A Categorical Explanatory Variable?

- Response variable (y): quantitative
- Have 1 categorical explanatory variable (x) with two categories.
- Model form:

$$y = \beta_0 + \beta_1 x + \epsilon$$

- First, need to convert the categories of x to numbers.

Example: Halloween Candy

```
1 candy <- read_csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-ranking/candy-data.csv")
2 glimpse(candy)
```

Rows: 85

Columns: 13

```
$ competitorname <chr> "100 Grand", "3 Musketeers", "One dime", "One quarter..."
$ chocolate      <dbl> 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,...
$ fruity         <dbl> 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 1,...
$ caramel        <dbl> 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0,...
$ peanutyalmondy <dbl> 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ nougat         <dbl> 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,...
$ crispedricewafer <dbl> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ hard           <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1,...
$ bar            <dbl> 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,...
$ pluribus       <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 0, 1,...
$ sugarpercent   <dbl> 0.732, 0.604, 0.011, 0.011, 0.906, 0.465, 0.604, 0.31...
$ pricepercent   <dbl> 0.860, 0.511, 0.116, 0.511, 0.511, 0.767, 0.767, 0.51...
$ winpercent     <dbl> 66.97173, 67.60294, 32.26109, 46.11650, 52.34146, 50...
```

What might be a good categorical explanatory variable of **winpercent**?

Exploratory Data Analysis

Before building the model, let's explore and visualize the data!

- What `dplyr` functions should I use to find the mean and sd of `winpercent` by the categories of `chocolate`?
- What graph should we use to visualize the `winpercent` scores by `chocolate`?

Exploratory Data Analysis

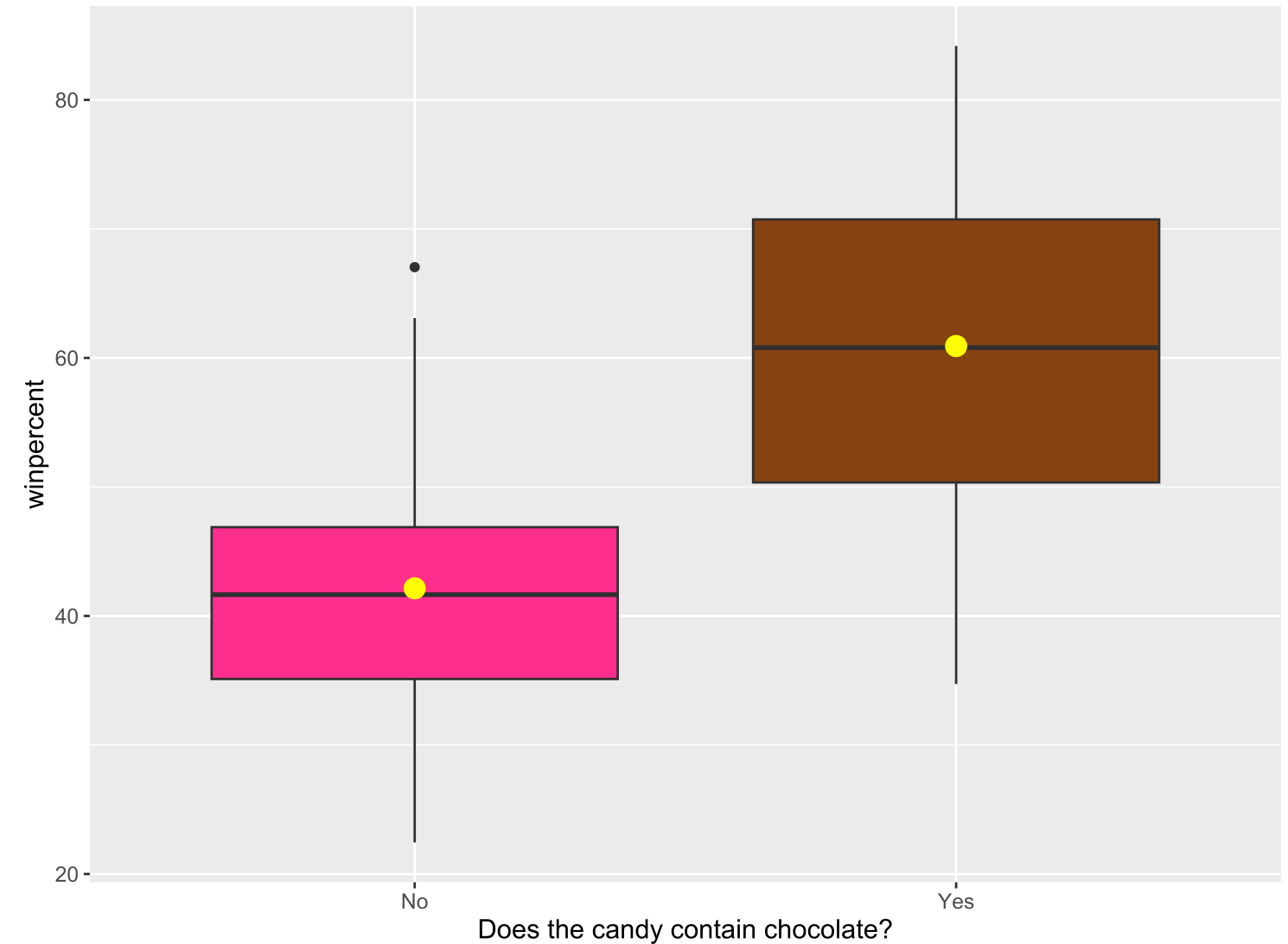
```
1 # Summarize
2 candy %>%
3   group_by(chocolate) %>%
4   summarize(count = n(), mean_win = mean(winpercent),
5             sd_win = sd(winpercent))
```

```
# A tibble: 2 × 4
```

	chocolate	count	mean_win	sd_win
	<dbl>	<int>	<dbl>	<dbl>
1	0	48	42.1	10.2
2	1	37	60.9	12.8

Exploratory Data Analysis

```
1 ggplot(candy, aes(x = factor(chocolate),
2                     y = winpercent,
3                     fill = factor(chocolate))) +
4   geom_boxplot() +
5     stat_summary(fun = mean,
6                 geom = "point",
7                 color = "yellow",
8                 size = 4) +
9   guides(fill = "none") +
10  scale_fill_manual(values =
11                    c("0" = "deeppink",
12                      "1" = "chocolate4")) +
13  scale_x_discrete(labels = c("No", "Yes"),
14                  name =
15    "Does the candy contain chocolate?")
```



Fit the Linear Regression Model

Model Form:

$$y = \beta_0 + \beta_1 x + \epsilon$$

When $x = 0$:

When $x = 1$:

```
1 mod <- lm(winpercent ~ chocolate, data = candy)
2 library(moderndive)
3 get_regression_table(mod)
```

A tibble: 2 × 7

	term	estimate	std_error	statistic	p_value	lower_ci	upper_ci
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	intercept	42.1	1.65	25.6	0	38.9	45.4
2	chocolate	18.8	2.50	7.52	0	13.8	23.7

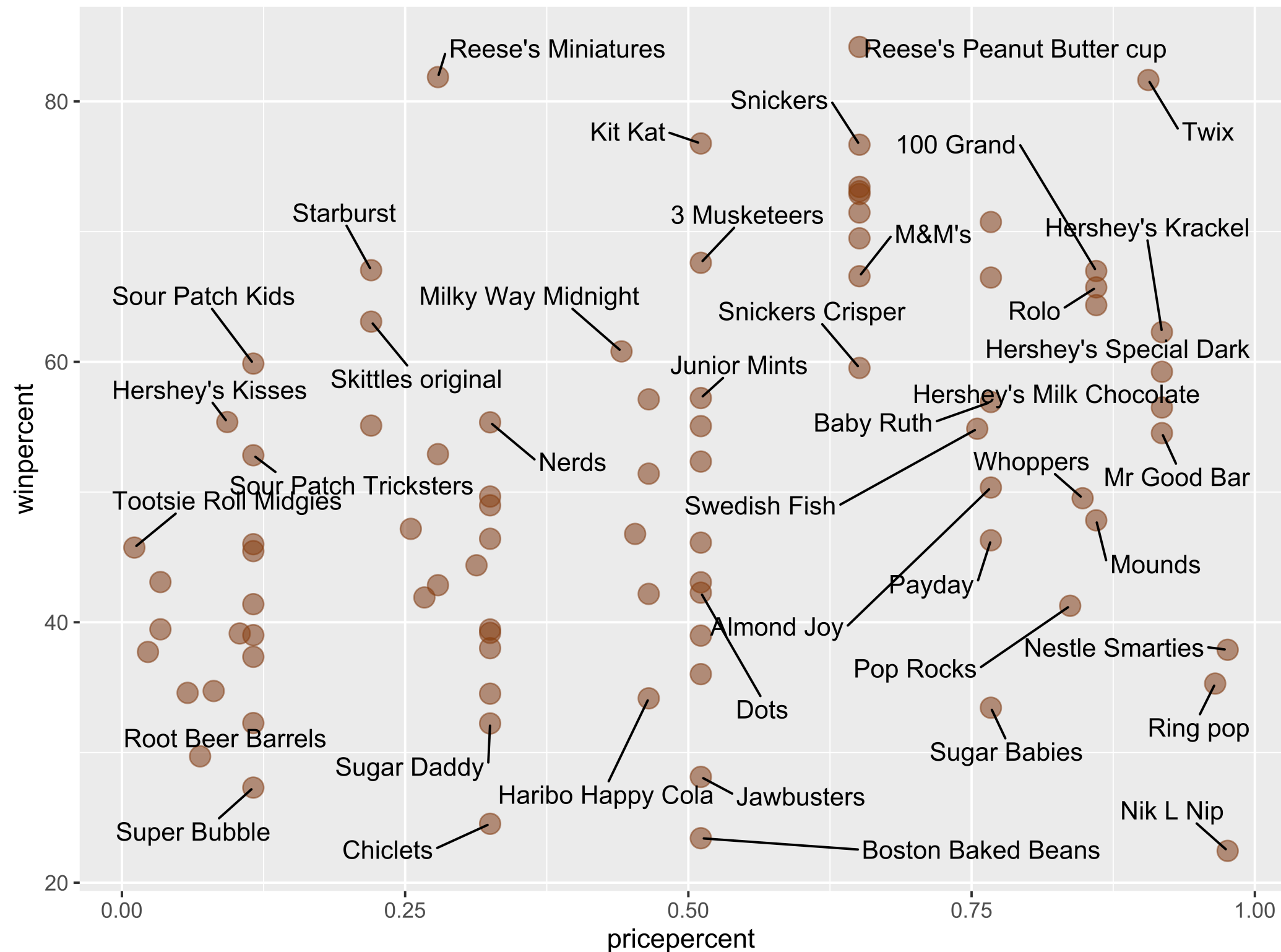
Notes

- When the explanatory variable is categorical, β_0 and β_1 no longer represent the intercept and slope.
- Now β_0 represents the (population) mean of the response variable when $x = 0$.
- And, β_1 represents the change in the (population) mean response going from $x = 0$ to $x = 1$.
- Can also do prediction:

```
1 new_candy <- data.frame(chocolate = c(0, 1))
2 predict(mod, newdata = new_candy)
```

```
      1      2
42.14226 60.92153
```


Turns Out Reese's Miniatures Are Under-Priced...



Note about Office Hours

- Professor McConville: Tuesdays 2:30 - 4pm and Wednesdays 2 - 3pm in Taylor 212.
- Rebecca: Mondays and Wednesdays 4:30 - 6pm in the classroom (Taylor 203).
- Going to add one more hour. Vote on the time: burl.live/ohs

